utility10_04-bijli-meter-reader

Bijli Meter Reader: manufacturing manual

> **Status: design only.** Nothing here has been built or tested as hardware. This manual tells you exactly what to make, buy, assemble, program and test, and how to accept or reject a unit; expect to correct it after the first prototype run (5 units).

Product summary. A clip-on photodiode head sits on the meter's pulse LED. The unit counts pulses (20 ms debounce), shows live watts, kWh since reset and an estimated bill from slab rates you enter. **No mains contact.**

Product line / price in the plan: Tier A · low-voltage electronics · prototype Rs 900 · price Rs 1299. Status: designed and verified as far as a computer can; no prototype built, nothing field-tested.

1. What you are building: deliverables

Deliverable	Where	State
PCB: Gerbers, BOM, CPL, KiCad project	`utility10/pcb/release/bijli/`	**order-ready: every automatic check passes** (DRC errors 0, netlist match True)
Firmware (`boards/bijli_meter.hex`)	`utility10/firmware/release/8MHz/boards/bijli_meter.hex`, source `utility10/firmware/src/main_board.cpp` (seeti: `seeti_prod.cpp`), drivers `board_drivers.h`, engine `utility10/firmware/lib/utilcore.c`	engine host-tested (115 checks); board firmware compiles but is **not yet run on hardware** (see section 6)
Enclosure STL (prototype shell)	`utility10/04-bijli-meter-reader/enclosure_body.stl`, `enclosure_lid.stl` and the board's `4_enclosure/`	prototype shell only
Mechanical / bought-in parts	section 3	listed
Test sheet	`manuals/templates/eol_test_sheet.md`	template

2. Before you start

- Read `manuals/STANDARD_PROCEDURES.md` (sections A-F, J, K).
- Tools: soldering iron, flux, multimeter, USBasp programmer + avrdude, calipers, 3D printer or print service, bench supply with current limit.
- Power source for this product: **5 V USB charger (not supplied with the PCB)**.

3. Bill of materials

3a. Circuit board parts (20 parts, 3 test points)

SMD parts marked `SMT (JLC)` are assembled by the fab from the JLC BOM; parts marked `hand/THT` are soldered by you. Every LCSC number was checked against an LCSC/JLCPCB listing on 2026-11-01; re-check stock on the day you order.

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
U1	ATmega328P-AU	TQFP-32_7x7mm_P0.8mm	ATMEGA328P-AU	C14877	SMT (JLC)
J1	USB-C 5V	USB_C_Receptacle_HRO_TYPE-C-31-M-12	TYPE-C-31-M-12	C165948	SMT (JLC)
F1	PTC 500mA	Fuse_1206_3216Metric	Littelfuse 1206L050YR (6 V)	C163512	SMT (JLC)
D3	SMAJ5.0A	D_SMA	Littelfuse SMAJ5.0A	C83329	SMT (JLC)
J2	ISP	PinHeader_2x03_P2.54mm_Vertical	ZHOURI 2.54-2*3	C5116479	hand/THT
J3	Sensor head	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
J4	TM1637 display	JST_PH_B4B-PH-K_1x04_P2.00mm_Vertical	JST B4B-PH-K-S(LF)(SN)	C131334	hand/THT
SW1	Reset bill	SW_SPST_B3U-1000P	Omron B3U-1000P	C231329	SMT (JLC)
D1	Green	LED_0603_1608Metric	Everlight 19-217/GHC-YR1S2/3T (LCSC showed it out of stock on the search day: check)	C72043	SMT (JLC)
R1	5.1k	R_0603_1608Metric	0603 5.1k 1%	C23186	SMT (JLC)
R2	5.1k	R_0603_1608Metric	0603 5.1k 1%	C23186	SMT (JLC)
R3	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
R4	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
R5	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
R6	100	R_0603_1608Metric	UNI-ROYAL 0603WAF1000T5E	C22775	SMT (JLC)
C1	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
C2	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C3	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C4	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)
C5	100n	C_0603_1608Metric	0603 100nF X7R	C14663	SMT (JLC)

3b. Bought-in and mechanical items (estimates for about 2,000 units; replace with quotes)

Item	Rs
MCU + crystal	45
Photodiode + amp	25
4-digit display	35
Clip head + cable	40
Battery/USB	30
PCB	30
Enclosure	50
Packaging	30
Assembly + test	40

4. Step 1: fabricate the PCB

Upload `utility10/pcb/release/bijli/1_fabrication/bijli_meter_reader_gerbers.zip` with the settings in STANDARD_PROCEDURES section A. Finish: HASL lead-free. Board size and layer list are in `utility10/pcb/release/bijli/README.md`.

5. Step 2: assemble the board

1. Order SMT assembly as in STANDARD_PROCEDURES section B (or hand-solder the SMD parts in section D order).
2. Solder the through-hole parts listed `hand/THT` in 3a.
3. Clean, inspect under magnification, and compare with `2_assembly/*_assembly_top.pdf`.

6. Step 3: program the microcontroller

1. Connect a USBasp to the 6-pin ISP header J2 (pin 1 = square pad) with the board powered (5 V USB charger (not supplied with the PCB)).
2. Set fuses: `avrdude -p m328p -c usbasp -U lfuse:w:0xE2:m -U hfuse:w:0xD9:m -U efuse:w:0xFD:m` (internal 8 MHz RC, CKDIV8 off, brown-out 2.7 V (the 8 MHz speed-grade limit): the unit resets cleanly when the battery is flat).
3. Flash and verify: `avrdude -p m328p -c usbasp -U flash:w:utility10/firmware/release/8MHz/boards/bijli_meter.hex:i` then `-U flash:v:utility10/firmware/release/8MHz/boards/bijli_meter.hex:i`.
4. Remove the programmer, power-cycle the board, and confirm the start-up behaviour (section 9).

If you change the supply type (for example USB instead of batteries) use the hex built for that clock (`4MHz` for 2 x AAA, `8MHz` for every other tool).

Firmware pin map (generated from the PCB spec, so firmware and board agree):

Arduino pin	ATmega pin	Net
D3	1	DCLK
D4	2	DIO
D7	11	LED
A0	23	SIG

Arduino pin	ATmega pin	Net
D2	32	BTN

What is verified: the control logic (`utilcore.c`) passes 115 host tests. **What is not:** the display, OLED, RTC and HX711 drivers and the button flows in `main_board.cpp` were written from datasheets and only compile-checked. Budget bring-up time for them on the first prototype.

7. Step 4: electrical bring-up

1. Plug a 5 V charger: VBUS (TP at J1/F1) = 5.0 V; +5V (TP1) = 4.9-5.0 V.
2. SIG with head disconnected = 5 V (pull-up); with the phototransistor in the dark about 4.9 V; under a lit meter LED a drop of at least 0.5 V.
3. Flash over ISP: `avrdude -p m328p -c usbasp -U lfuse:w:0xE2:m -U hfuse:w:0xD9:m -U efuse:w:0xFD:m` (internal 8 MHz, brown-out 2.7 V; USB supply is 5 V).
4. Hold the head over a 1 Hz test LED: display counts one pulse per second.

8. Step 5: sensors, cables and mechanical assembly

- **Clip-on sensor head.** Solder the phototransistor/photodiode (J3 3-pin lead: +5 V, SIG, GND) on a 1.5 m cable inside a black ABS clip that blocks ambient light and sits over the meter's pulse LED; use heat-shrink at the cable exit. **There is no mains contact: never open the meter or touch its terminals.**
- Display: a TM1637 4-digit module plugs into the 4-pin JST-PH J4 (GND, +5V, DIO, CLK).

9. Step 6: enclosure and final assembly

1. Print the shell (STANDARD_PROCEDURES section G). Enclosure: `enclosure_body.stl` / `enclosure_lid.stl` (80×50×22 mm, 2.4 mm walls, watertight): a simple prototype shell; production tooling needs a proper design.
2. Place the board on the standoffs (4 mm), fasten with M2.5 screws (6 mm), route cables through the cut-outs, close the lid.
3. Label: serial number, rev, battery type/polarity, warnings.

10. Step 7: calibration

- Enter the meter constant (pulses per kWh, printed on the meter) in the setup; verify with a known load (a 100 W lamp for 10 minutes gives about 0.017 kWh).

11. Step 8: end-of-line functional test (every unit)

1. Flash a 1 Hz LED at the head: count = 1 per second; cover with ambient light: no spurious counts.
2. Check that watts and kWh are displayed and the estimated bill uses the slab rates entered.

Record results on `manuals/templates/eol_test_sheet.md`. Accept only if every line passes.

12. Packaging, documents, compliance

- In the box: unit, batteries or note on the required adapter, mounting accessories, quick-start card (`manuals/templates/quick_start_card.md`), warranty card.
- Compliance items to verify before sale: Battery safety, BIS requirement for any bought-in adapter, EMC, e-waste EPR, Legal Metrology labelling; no safety claims..

- Labelling: STANDARD_PROCEDURES K.

13. Troubleshooting (product specific)

Symptom	Cause / action
No pulses	head not aligned to the pulse LED: check SIG voltage swing, at least 0.5 V drop when the LED is lit
Count too high	ambient light leak: improve the clip shielding

General issues: STANDARD_PROCEDURES section L.

14. Known limits and risks

Host tests: 1000 W for one simulated hour produces 1600 pulses and 1.000 kWh; watts from the pulse gap within 4%; idle shows 0 W; slab bill arithmetic; bounce rejected. Slab rates in the code are placeholders, **not a real DISCOM tariff**. Not verified: sensing under sunlight, different LED colours/meter types.

Meter types differ; some new smart meters have no accessible LED; customers may expect an app.

Nothing in this manual replaces testing with real users, a design review by a PCB engineer, and the compliance checks above.